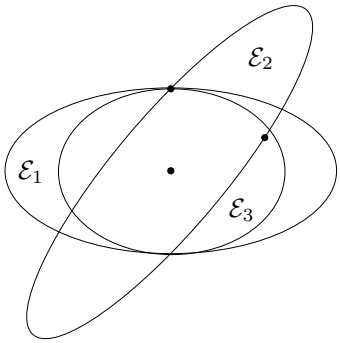


Figure 2.18 Three ellipsoids in $\mathbf{R}^2$, centered at the origin (shown as the lower dot), that contain the points shown as the upper dots. The ellipsoid $\mathcal{E}_1$ is not minimal, since there exist ellipsoids that contain the points, and are smaller (*e.g.*, $\mathcal{E}_3$). $\mathcal{E}_3$ is not minimal for the same reason. The ellipsoid $\mathcal{E}_2$ is minimal, since no other ellipsoid (centered at the origin) contains the points and is contained in $\mathcal{E}_2$.

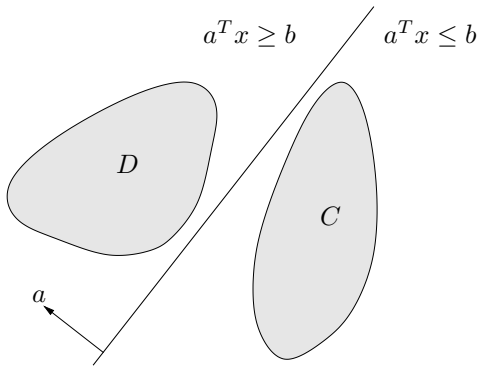


Figure 2.19 The hyperplane $\{x \mid a^T x = b\}$ separates the disjoint convex sets C and D. The affine function $a^T x - b$ is nonpositive on C and nonnegative on D.

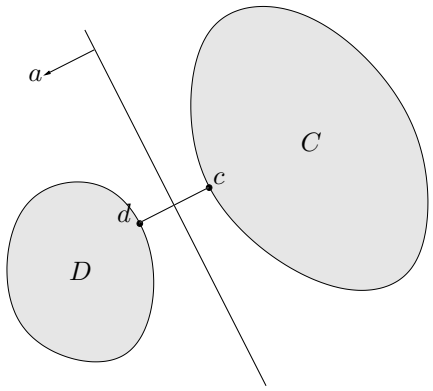


Figure 2.20 Construction of a separating hyperplane between two convex sets. The points $c \in C$ and $d \in D$ are the pair of points in the two sets that are closest to each other. The separating hyperplane is orthogonal to, and bisects, the line segment between c and d.

is positive, and that there exist points $c \in C$ and $d \in D$ that achieve the minimum distance, *i.e.*, $\|c - d\|_2 = \mathbf{dist}(C, D)$. (These conditions are satisfied, for example, when C and D are closed and one set is bounded.)

Define

$$a = d - c, \qquad b = \frac{\|d\|_2^2 - \|c\|_2^2}{2}.$$

We will show that the affine function

$$f(x) = a^T x - b = (d - c)^T (x - (1/2)(d + c))$$

is nonpositive on C and nonnegative on D, *i.e.*, that the hyperplane $\{x \mid a^T x = b\}$ separates C and D. This hyperplane is perpendicular to the line segment between c and d, and passes through its midpoint, as shown in figure 2.20.

We first show that f is nonnegative on D. The proof that f is nonpositive on C is similar (or follows by swapping C and D and considering $-f$). Suppose there were a point $u \in D$ for which

$$f(u) = (d - c)^T (u - (1/2)(d + c)) < 0. \tag{2.16}$$

We can express $f(u)$ as

$$f(u) = (d - c)^T (u - d + (1/2)(d - c)) = (d - c)^T (u - d) + (1/2)\|d - c\|_2^2.$$

We see that (2.16) implies $(d - c)^T (u - d) < 0$. Now we observe that

$$\frac{d}{dt}\|d + t(u - d) - c\|_2^2 \Big|_{t=0} = 2(d - c)^T (u - d) < 0,$$

so for some small $t > 0$, with $t \leq 1$, we have

$$\|d + t(u - d) - c\|_2 < \|d - c\|_2,$$

i.e., the point $d + t(u - d)$ is closer to c than d is. Since D is convex and contains d and u, we have $d + t(u - d) \in D$. But this is impossible, since d is assumed to be the point in D that is closest to C.

Example 2.19 *Separation of an affine and a convex set.* Suppose C is convex and D is affine, *i.e.*, $D = \{Fu + g \mid u \in \mathbf{R}^m\}$, where $F \in \mathbf{R}^{n\times m}$. Suppose C and D are disjoint, so by the separating hyperplane theorem there are $a \neq 0$ and b such that $a^T x \leq b$ for all $x \in C$ and $a^T x \geq b$ for all $x \in D$.

Now $a^T x \geq b$ for all $x \in D$ means $a^T F u \geq b - a^T g$ for all $u \in \mathbf{R}^m$. But a linear function is bounded below on $\mathbf{R}^m$ only when it is zero, so we conclude $a^T F = 0$ (and hence, $b \leq a^T g$).

Thus we conclude that there exists $a \neq 0$ such that $F^T a = 0$ and $a^T x \leq a^T g$ for all $x \in C$.

Strict separation

The separating hyperplane we constructed above satisfies the stronger condition that $a^T x < b$ for all $x \in C$ and $a^T x > b$ for all $x \in D$. This is called *strict separation* of the sets C and D. Simple examples show that in general, disjoint convex sets need not be strictly separable by a hyperplane (even when the sets are closed; see exercise 2.23). In many special cases, however, strict separation can be established.

Example 2.20 *Strict separation of a point and a closed convex set.* Let C be a closed convex set and $x_0 \notin C$. Then there exists a hyperplane that strictly separates x_0 from C.

To see this, note that the two sets C and $B(x_0, \epsilon)$ do not intersect for some $\epsilon > 0$. By the separating hyperplane theorem, there exist $a \neq 0$ and b such that $a^T x \leq b$ for $x \in C$ and $a^T x \geq b$ for $x \in B(x_0, \epsilon)$.

Using $B(x_0, \epsilon) = \{x_0 + u \mid \|u\|_2 \leq \epsilon\}$, the second condition can be expressed as

$$a^T(x_0 + u) \geq b \quad \text{for all} \quad \|u\|_2 \leq \epsilon.$$

The u that minimizes the lefthand side is $u = -\epsilon a/\|a\|_2$; using this value we have

$$a^T x_0 - \epsilon\|a\|_2 \geq b.$$

Therefore the affine function

$$f(x) = a^T x - b - \epsilon\|a\|_2/2$$

is negative on C and positive at x_0.

As an immediate consequence we can establish a fact that we already mentioned above: a closed convex set is the intersection of all halfspaces that contain it. Indeed, let C be closed and convex, and let S be the intersection of all halfspaces containing C. Obviously $x \in C \Rightarrow x \in S$. To show the converse, suppose there exists $x \in S$, $x \notin C$. By the strict separation result there exists a hyperplane that strictly separates x from C, *i.e.*, there is a halfspace containing C but not x. In other words, $x \notin S$.

Converse separating hyperplane theorems

The converse of the separating hyperplane theorem (*i.e.*, existence of a separating hyperplane implies that C and D do not intersect) is not true, unless one imposes additional constraints on C or D, even beyond convexity. As a simple counterexample, consider $C = D = \{0\} \subseteq \mathbf{R}$. Here the hyperplane $x = 0$ separates C and D.

By adding conditions on C and D various converse separation theorems can be derived. As a very simple example, suppose C and D are convex sets, with C open, and there exists an affine function f that is nonpositive on C and nonnegative on D. Then C and D are disjoint. (To see this we first note that f must be negative on C; for if f were zero at a point of C then f would take on positive values near the point, which is a contradiction. But then C and D must be disjoint since f is negative on C and nonnegative on D.) Putting this converse together with the separating hyperplane theorem, we have the following result: any two convex sets C and D, at least one of which is open, are disjoint if and only if there exists a separating hyperplane.

Example 2.21 *Theorem of alternatives for strict linear inequalities.* We derive the necessary and sufficient conditions for solvability of a system of strict linear inequalities

$$Ax \prec b. \tag{2.17}$$

These inequalities are infeasible if and only if the (convex) sets

$$C = \{b - Ax \mid x \in \mathbf{R}^n\}, \qquad D = \mathbf{R}^m_{++} = \{y \in \mathbf{R}^m \mid y \succ 0\}$$

do not intersect. The set D is open; C is an affine set. Hence by the result above, C and D are disjoint if and only if there exists a separating hyperplane, *i.e.*, a nonzero $\lambda \in \mathbf{R}^m$ and $\mu \in \mathbf{R}$ such that $\lambda^T y \leq \mu$ on C and $\lambda^T y \geq \mu$ on D.

Each of these conditions can be simplified. The first means $\lambda^T(b - Ax) \leq \mu$ for all x. This implies (as in example 2.19) that $A^T\lambda = 0$ and $\lambda^T b \leq \mu$. The second inequality means $\lambda^T y \geq \mu$ for all $y \succ 0$. This implies $\mu \leq 0$ and $\lambda \succeq 0$, $\lambda \neq 0$.

Putting it all together, we find that the set of strict inequalities (2.17) is infeasible if and only if there exists $\lambda \in \mathbf{R}^m$ such that

$$\lambda \neq 0, \qquad \lambda \succeq 0, \qquad A^T\lambda = 0, \qquad \lambda^T b \leq 0. \tag{2.18}$$

This is also a system of linear inequalities and linear equations in the variable $\lambda \in \mathbf{R}^m$. We say that (2.17) and (2.18) form a pair of *alternatives*: for any data A and b, exactly one of them is solvable.

2.5.2 Supporting hyperplanes

Suppose $C \subseteq \mathbf{R}^n$, and x_0 is a point in its boundary $\mathbf{bd}\, C$, *i.e.*,

$$x_0 \in \mathbf{bd}\, C = \mathbf{cl}\, C \setminus \mathbf{int}\, C.$$

If $a \neq 0$ satisfies $a^T x \leq a^T x_0$ for all $x \in C$, then the hyperplane $\{x \mid a^T x = a^T x_0\}$ is called a *supporting hyperplane* to C at the point x_0. This is equivalent to saying